Gospel Reading: **Mark 11:11-26**

And he entered Jerusalem, and went into the temple; and when he had looked round at everything, as it was already late, he went out to Bethany with the twelve. On the following day, when they came from Bethany, he was hungry. And seeing in the distance a fig tree in leaf, he went to see if he could find anything on it. When he came to it, he found nothing but leaves, for it was not the season for figs. And he said to it, "May no one ever eat fruit from you again." And his disciples heard it.
And they came to Jerusalem. And he entered the temple and began to drive out those who sold and those who bought in the temple, and he overturned the tables of the money-changers and the seats of those who sold pigeons; and he would not allow any one to carry anything through the temple. And he taught, and said to them, "Is it not written, `My house shall be called a house of prayer for all the nations'? But you have made it a den of robbers." And the chief priests and the scribes heard it and sought a way to destroy him; for they feared him, because all the multitude was astonished at his teaching.

Gospel Reading: **Mark 11:11-26**

And when evening came they went out of the city. As they passed by in the morning, they saw the fig tree withered away to its roots. And Peter remembered and said to him, "Master, look! The fig tree which you cursed has withered." And Jesus answered them, "Have faith in God. Truly, I say to you, whoever says to this mountain, `Be taken up and cast into the sea,' and does not doubt in his heart, but believes that what he says will come to pass, it will be done for him. Therefore I tell you, whatever you ask in prayer, believe that you have received it, and it will be yours. And whenever you stand praying, forgive, if you have anything against any one; so that your Father also who is in heaven may forgive you your trespasses. But if you do not forgive, neither will your Father who is in heaven forgive your trespasses"

Prayer

Lord Jesus, increase my faith and make my fruitful and effective in serving you and bringing you honor and glory in all that I do. Help me to be merciful and forgiving towards others just as you have been merciful and forgiving towards me.

St. John Baptist De La Salle
Pray for us
Live Jesus in our hearts
Forever

Attendance

Project

- 20 minute presentation and 10 minute Q & A
- The order of presentation will be according to group no.
- Submit the Document (Include RE, DFA, CFG, PDA, User’s Manual, and Sample Outputs) and the app link.

Context-Free Grammar

5. $S \rightarrow SS \mid BS \mid SB \mid USU \mid \Lambda$

$B \rightarrow aa \mid bb$

$U \rightarrow ab \mid ba$

CFL :

Context-Free Grammar

7. All strings in which the nos. of a is even.

Context-Free Grammar

8. All strings without substring ba.

Pushdown Automata

Learning Outcomes

1. Recognize the components of a pushdown automaton particularly its use of an infinite stack.
2. Identify the languages accepted by given PDAs.
3. Design PDAs of given languages.

Pushdown automata

- is a finite automata with extra memory called stack which helps Pushdown automata to recognize Context Free Languages.
- is a way to implement a context-free grammar in a similar way we design DFA for a regular grammar.
- can remember an infinite amount of information.

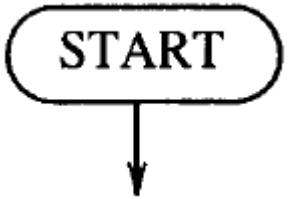
DEFINITION

A pushdown automaton, PDA, is a collection of eight things:

1. An alphabet **Σ** of input letters.
2. An input TAPE (infinite in one direction). Initially the string of input letters is placed on the TAPE starting in cell i. The rest of the TAPE is blank.
3. An alphabet Γ of STACK characters.

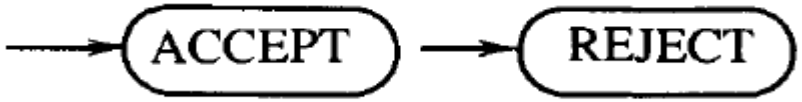
DEFINITION

4. A pushdown STACK (infinite in one direction). Initially the STACK is empty (contains all blanks).
5. One START state that has only out-edges, no in-edges.



DEFINITION

6. Halt states of two kinds: some ACCEPT and some REJECT. They have in-edges and no out-edges.

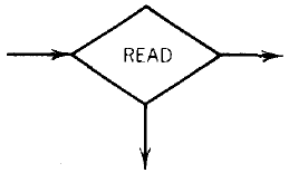


7. Finitely many nonbranching PUSH states that introduce characters onto the top of the STACK. They are of the form

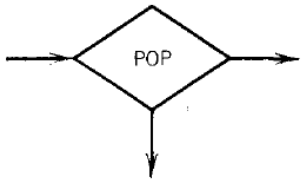
DEFINITION

8. Finitely many branching states of two kinds:

(i) States that read the next unused letter from the TAPE



(ii) States that read the top character of the STACK



Examples

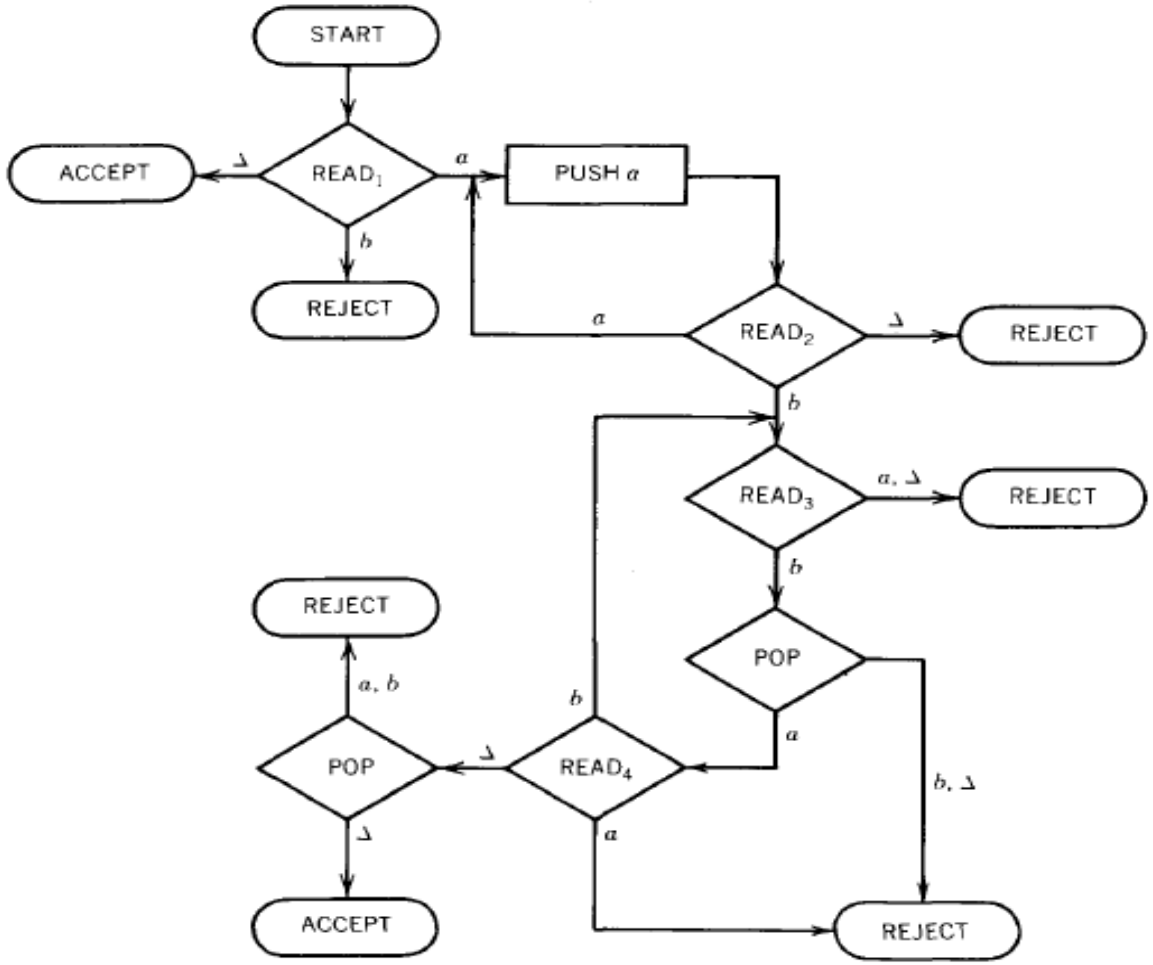
1. All strings that start with a.

2. All strings that end with a.

3. All strings that start with double letter.

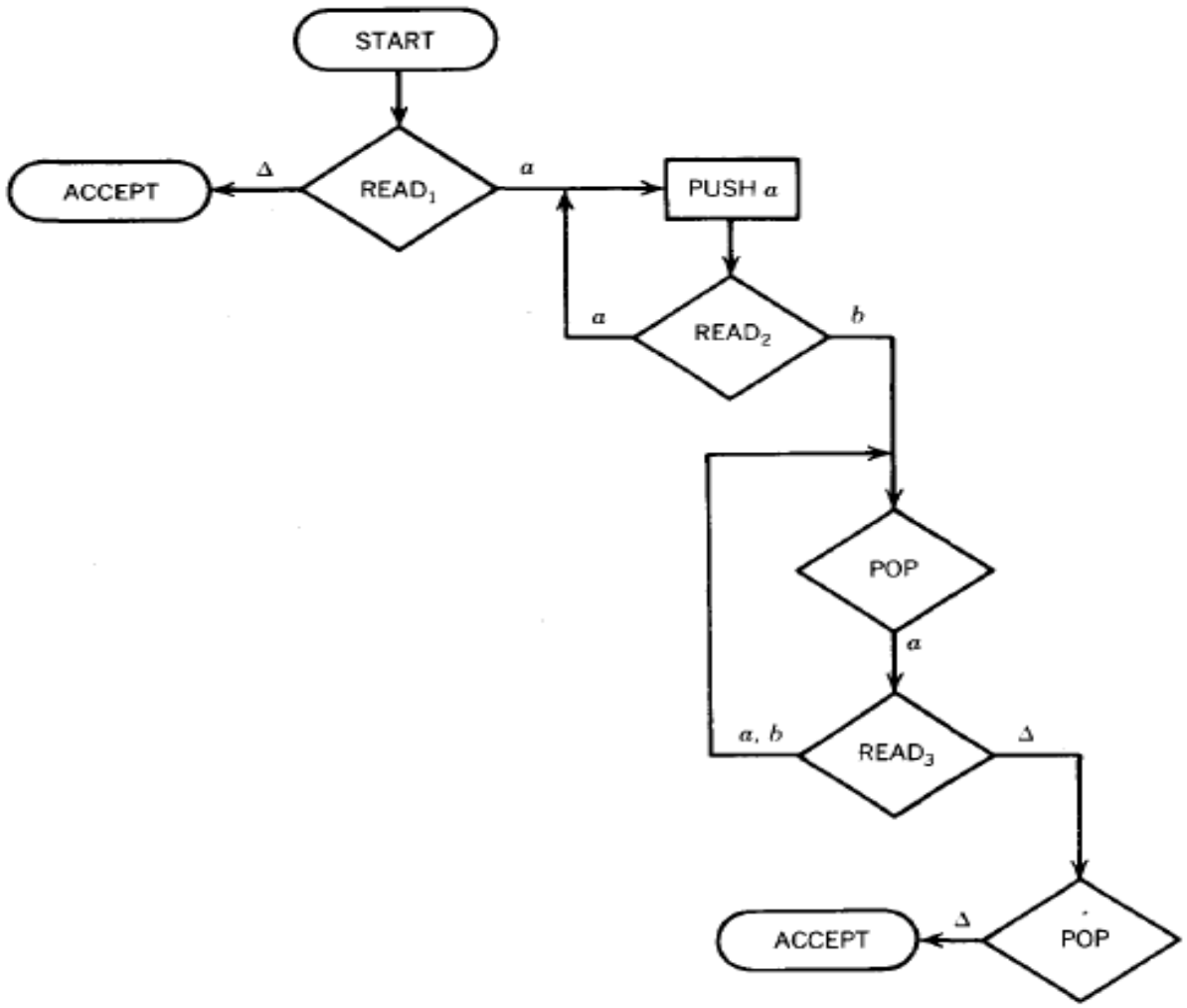
4. All strings that end with double letter.

examples



(i) *abb*
(ii) *abab*
(iii) *aabb*
(iv) *aabbbb*

examples



Trace the following words on this PDA, write whether valid or not:
(i) *aaabbb*
(ii) *aaabab*
(iii) *aaabaa*
(iv) *aaaabb*
(v) aaabbba

Identify the language being accepted by this PDA: (5 pts.)

references

https://www.geeksforgeeks.org/introduction-of-pushdown-automata/

https://www.tutorialspoint.com/automata_theory/pushdown_automata_introduction.htm

Introduction to Computer Theory by Daniel Cohen